

Revision 1: This report is revised to include the results of the stability retool and updated scope for the Physical Interconnection of the facility. The Sunbury – Dauphin 230kV network reinforcement that was previously identified is no longer needed. In addition, relay modification work for the Physical Interconnection is not required. All references to these scope of work have been deleted.

Generation Interconnection System Impact Study Report

For

***PJM Generation Interconnection Request
Queue Position AC2-092***

***Susquehanna-Lackawanna 500kV
65 MW Capacity / 55 MW Energy***

February 2022

Introduction

This System Impact Study (SIS) has been prepared in accordance with the PJM Open Access Transmission Tariff, Section 205, as well as the System Impact Study Agreement between Moxie Freedom LLC, the Interconnection Customer (IC) and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is PPL Electric Utilities (PPL EU).

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the IC. As a requirement for interconnection, the IC may be responsible for the cost of constructing Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an IC may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The IC is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

Moxie Freedom LLC, the Interconnection Customer (IC), has proposed an increase of 55 MWE (65 MWC) to its natural gas generating facility located in the Luzerne County of Pennsylvania, which is part of the PPL Electric Utilities (PPL EU) Susquehanna Region. Under the original queue position Z2-046, the IC proposed 1050 MWE (900 MWC). Under a subsequent queue position, AA1-066, the IC proposed an increase of 0 MWE (80 MWC). The proposed combined capability of queue positions Z2-046, AA1-066 and AC2-092 is 1105 MW with 1045 MW being recognized by PJM as Capacity.

The previous queues and uprates are summarized below:

Queue	Maximum Facility Output (MW)	Energy (MW)	Capacity (MW)
Z2-046	1050	1050	900
AA1-066 (Proposed Increase)	0	0	80
AC2-092 (Proposed Increase)	55	55	65
Z2-046/AA1-066/AC2-092 Total	1105	1105	1045

The queue position AC2-092 planned in-service date is June 2021. **This study does not imply a PPL EU commitment to this in-service date.**

Point of Interconnection

The IC requested a transmission level interconnection. As a result, the queue position AC2-092 will interconnect with the PPL EU transmission network via the Susquehanna-Shickshinny and Shickshinny-Lackawanna 500kV lines (formerly the Susquehanna-Lackawanna 500kV line) through the Shickshinny 500kV switchyard. The queue position AC2-092 POI is identical to the queue position Z2-046 POI. See Attachment 1 of this Study for more information. Also refer to queue positions Z2-046 and AA1-066 on the PJM website for more information.

Cost Summary

The AC2-092 interconnection request will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$ 0
Direct Connection Network Upgrades	\$ 0
Non Direct Connection Network Upgrades	\$ 0
Total Costs	\$ 0

In addition, the AC2-092 project may be responsible for a contribution to the following costs:

Description	Total Cost
New System Upgrades	\$ 0
Previously Identified Upgrades	\$ 0
Total Costs	\$ 0

Cost allocations for these upgrades will be provided in the Facilities Study, if necessary.

Attachment Facilities

None

Direct Connection Cost Estimate

None.

Non-Direct Connection Cost Estimate

None.

Estimated Schedule

The estimated time is **0** months from engineering start to construction finish, after the PJM three-party Interconnection Service Agreement (ISA) is signed.

Assumptions

- No major environmental, real estate, or permitting issues are anticipated.
- Suitable line/equipment outages can be scheduled as required. Failure to meet a scheduled facility outage may result in project delays.
- During construction, if extreme weather conditions or other system safety concerns arise, field construction may need to be rescheduled, which could possibly delay the schedule.
- No load changes on auxiliary transformers.
- In the event of any operational, governmental, and/or environmental regulatory delays, the use of additional resources, such as overtime, premiums for expedited material, and/or contractor labor, may enable PPL EU to decrease this construction period but no guarantees can be made.
- If a Facilities Study is required, the estimated PPL EU duration and cost estimate to perform the study is 7 months; It is estimated to cost \$50,000.

Interconnection Customer Requirements

IC Point of Interconnection (POI)

The AC2-092 POI is identical to the Z2-046 POI.

IC Voice Communication Circuit Requirements

The IC Voice Communication requirements are identical to those specified under Z2-046.

IC Protective Relaying Requirements

The IC Protective Relaying requirements are identical to those specified under Z2-046.

IC Substation Intertie Protective Relaying (IPR) and Point of Contact (POC) Fault Interrupting Device (FID) Requirements

The IC Substation IPR and POC FID requirements are identical to those specified under Z2-046.

Refer to the website addresses shown below for the PPL EU IPR and POC requirements:

IPR Requirements

<http://www.pjm.com/-/media/planning/plan-standards/private-ppl/parallel-generation-requirements.ashx?la=en>

POC Requirements

<http://www.pjm.com/-/media/planning/plan-standards/private-ppl/point-of-contact-requirements.ashx?la=en>

IC Generator Harmonic and Flicker Requirements

The IC Generator Harmonic and Flicker requirements are identical to those specified under Z2-046.

IC Generator Regulation or Reactive Support Requirements

The IC Generator Regulation or Reactive Support requirements are identical to those specified under Z2-046.

IC Generator Voltage Schedule Requirements

The IC Generator Voltage Schedule requirements are identical to those specified under Z2-046.

IC Distribution Service Requirements

The IC Distribution Service requirements are identical to those specified under Z2-046.

IC PA PUC Certification & Environmental Requirements

The IC PA PUC Certification & Environmental requirements are identical to those specified under Z2-046.

IC Revenue Metering and SCADA Requirements

The IC Protective Relaying requirements are identical to those specified under Z2-046.

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

Interconnected Transmission Owner Requirements

The IC Protective Relaying requirements are identical to those specified under Z2-046.

Network Impacts

The Queue Project AC2-092 was evaluated as a 65.0 MW (Capacity 65.0 MW) injection at on the Lackawanna – Susquehanna 500kV line in the PPL area. Project AC2-092 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AC2-092 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis – 2020

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None.

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output).

None.

Short Circuit

(Summary of impacted circuit breakers)

New circuit breakers found to be over-duty:

None.

Contributions to previously identified circuit breakers found to be over-duty:

None

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this interconnection request)

None.

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None.

Steady-State Voltage Requirements

(Summary of the VAR requirements based upon the results of the steady-state voltage studies)

None.

Stability and Reactive Power Requirement for Low Voltage Ride Through

(Summary of the VAR requirements based upon the results of the dynamic studies)

None.

Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The IC can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this interconnection request by addressing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

Not Applicable.

Light Load Analysis in 2020

Not Required.

Affected System Analysis & Mitigation

NYISO Impacts:

None.

Attachment 1.

System Configuration

